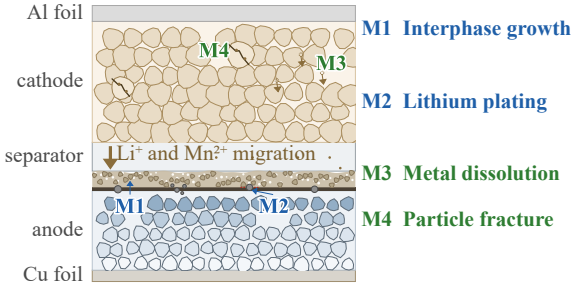
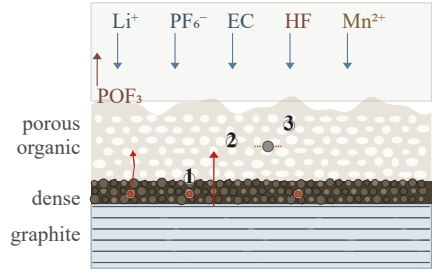
(a) Locus of each degradation mechanism



(b) Architecture of the interphase



sublayer ≈ 4 nm; compact film 20–50 nm

(c) Coupling topology of the four mechanisms



(d) Closure condition of each loop



R_SEI > R*, R* = (Δ − η_ct)/j

Mn loss > Li loss

interphase vol > pore vol

salt use > salt supply

Each loop reinforces itself; the first to reach its closure condition sets the limit.

Green denotes the sources M3 and M4, which generate reactive surface; blue denotes the sinks M1 and M2, which consume it and raise η. Drawings are schematic: no dimension is implied for particle, island or dendrite size.